

Raphael Sena

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PROFILE

Backend-focused Software Engineer working across two concurrent **Java/Spring Boot** codebases: production ticketing systems serving 177,000+ users at Modaxo, and, as a freelance backend developer, a multi-tenant **ERP/tax** platform built from scratch with **Java 21**, **PostgreSQL**, **Flyway** and **event-driven architecture**. Comfortable owning a feature from data model and **JPA/Hibernate** persistence through **REST API** design, testing and release. Software Engineering student at PUC Minas. English C1, Spanish A2.

EXPERIENCE

- **New Energy Soluções Elétricas** Belo Horizonte, MG - Brazil (Remote)
Backend Developer (Freelance) Mar 2026 – Present
 - **Precifique – Pricing & Tax Engine (from scratch)**: Built a commercial pricing and Brazilian tax calculation system from the ground up on **Java 21**, **Spring Boot 3.2**, **PostgreSQL** and **Flyway** migrations, replacing a legacy Excel-based workflow. Modeled NCM-based fiscal classification via **JPA/Hibernate** entity relationships and integrated the **IBPT** tax-data API.
 - **Kvaris – Multi-Tenant ERP Platform**: Evolved Precifique into a broader **multi-tenant ERP** serving two legal entities on a shared database, with tenant isolation via a request-scoped **TenantContext/TenantInterceptor**. Architected the Sales/Purchasing/WMS/Fiscal/Finance pipeline as a **Modular Monolith** with internal **event-driven** communication (**Spring ApplicationEventPublisher**) and a dedicated state machine owning order-status transitions.
 - **Auth & Integrations**: Implemented **OAuth2/OIDC** single sign-on via Microsoft Entra ID (Azure AD) with role-based App Roles, and scheduled **third-party API** sync jobs against external ERP systems for both tenant companies.
 - **Engineering Judgment**: Made deliberate architecture trade-offs to avoid over-engineering: chose a cost-effective VPS/PaaS deployment over managed cloud services, and a simplified internal Outbox pattern over a Kafka/Debezium event pipeline. Currently in active development on staging, demoable live.
- **Modaxo** Belo Horizonte, MG - Brazil
Software Engineer May 2025 – Present
 - **Sigom Cloud – Payment Reporting (Java)**: Rebuilt **Java/Spring Boot** payment report generation on a revenue management platform, cutting estimated computational effort from **20M–1T to 1M** operations at high data volumes, and delivered the supporting **REST** endpoints end to end.
 - **Corrective & Evolutionary Maintenance (SIGO)**: Develop and maintain **Java/Spring Boot** services behind an electronic ticketing product deployed across 47 cities with **695K+ downloads**: investigate and resolve production incidents, evolve features, and maintain **REST** integrations on a distributed, service-oriented back end.
 - **Data Integrity Across Systems**: Automated database versioning and migrations across **120 production systems** using **Liquibase**, eliminating manual **SQL** drift and cutting deployment risk for legacy **Java** services.
 - **Ways of Working**: Work in **Scrum** squads: technical refinements and estimates, **code review**, **Jira/Confluence** tracking and documentation, and incremental delivery alongside product stakeholders.
- **AVASO Technology Solutions** Belo Horizonte, MG - Brazil
Field Support Engineer Sep 2023 – May 2025
 - **Service Desk (Freelance)**: Resolved hardware and software incidents across desktops, laptops, VMs, servers, backup systems, VoIP devices, and peripherals, sustaining **95% SLA compliance** for a multicultural English-speaking user base, working as an allocated engineer for international clients.
- **Sociedade Mineira de Cultura (PUC Minas' governing foundation)** Belo Horizonte, MG - Brazil
IT Technician Sep 2021 – Sep 2023
 - **Telephony Migration**: Drove a VoIP migration for **100+ users**, reconfiguring logical network topology and deploying endpoint hardware during cutover.
 - **Service Desk & Help Desk**: Delivered L1 technical support using Active Directory, ticketing systems (CSC), VPN, and directory/printer mapping platforms under established SLA standards.

EDUCATION

- **Pontifical Catholic University of Minas Gerais** Belo Horizonte, MG - Brazil
Bachelor of Software Engineering *Jul 2023 – Jul 2027*
 - **Experimental Software Agency - PMMG: Tech Lead** for the HR recruitment platform of the Military Police of Minas Gerais, serving **40,000+** candidates per examination cycle, owning scoping through delivery.
 - **2x Interdisciplinary Project Champion:** Led two award-winning teams delivering production software for external clients.
 - **Research Project:** Member of a research group advised by Prof. Lucila Ishitani, publishing a paper on productivity in software engineering.
- **Kogarah High School** Sydney, NSW - Australia
High School Exchange Program *Jul 2017 – Dec 2017*

PROJECTS

- **Remediar – NGO ERP:**  
 - **Featured Project:** Award-winning **ERP** platform built and deployed for NGO Remediar, an organization focused on medicine donation, covering inventory, donations and reporting.
 - **Technologies:** Java, Spring Boot, **REST APIs**, **Microservices**, PostgreSQL, Docker, Nginx, Linux, **CI/CD**, TypeScript, Next.js, React.
- **Rural ERP with AI Copilot (Final Thesis):**
 - **Capstone Project:** Modular **ERP** for small-scale producers with an irrigation control module, designed with **DDD** and validated through **ATAM**. Includes an **LLM** agronomist copilot with **RAG** over public agricultural datasets and anomaly detection over sensor data.
 - **Technologies:** Java, Spring Boot (modular), PostgreSQL + pgvector, MQTT, Go edge gateway, Flutter, Docker Compose, AWS.

SKILLS

- **Backend:** Java 8–21, Spring Boot, Spring Framework, Spring Security, **REST APIs**, **JPA/Hibernate**, Node.js, C#.
- **Architecture:** **Microservices**, Modular Monolith, **Event-Driven Architecture** (Spring ApplicationEventPublisher), Domain-Driven Design, Multi-Tenancy, State Machines, Distributed Systems, System Integration.
- **Databases:** **PostgreSQL**, **Flyway**, Oracle, MySQL, SQLite, Liquibase.
- **Auth & Integration:** OAuth2/OIDC, Microsoft Entra ID (Azure AD SSO), third-party API integrations, Firebase.
- **DevOps & Cloud:** Docker, Git, pull-request code review, **CI/CD**, GitHub Actions, Maven, AWS (EC2, Lightsail), Google Cloud (BigQuery), Linux.
- **Testing & Quality:** JUnit, Mockito, Automated Testing, Code Review, SonarQube.
- **Observability & Messaging:** Grafana, Prometheus, Zipkin, Crashlytics, RabbitMQ, Redis.
- **Domain:** Fintech/Tax (Brazilian tax: ICMS, IPI, PIS/COFINS, NCM), ERP, SaaS, AI-enabled products.
- **Practices:** Scrum, Kanban, Jira, Confluence, Git Flow, remote collaboration.

LANGUAGES

- **English** (C1 - Advanced) | **Portuguese** (Native) | **Spanish** (A2 - Elementary) | **German** (B1 - Intermediate)